# Joseph Li

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## SKILLS

**Languages & Databases:** Python, Go, C, C++, CUDA, TypeScript, JavaScript, SQL, PostgreSQL, Redis

**Infrastructure & Development:** Linux, Docker, Kubernetes, Git, GitHub Actions, QEMU, AWS Bedrock, Next.js, React, Vercel

**Systems & AI:** API Gateways, CI/CD, Regression Testing, Observability, Asynchronous Programming, Operating Systems, Virtual Memory, Langfuse, LLM Evaluation, PyTorch, vLLM, Roofline Analysis, TensorFlow, LoRA, scikit-learn, NumPy, Pandas

## PROFESSIONAL EXPERIENCE

**CORTEX AI** **Menlo Park, CA**

***Software Engineer Intern*** ***Jun. 2026 - Aug. 2026***

- Designed and launched an LLM API gateway in **Go**, **PostgreSQL**, and **Redis** that routes OpenAI, Anthropic, and **AWS Bedrock** through one endpoint for a 2,000-seat customer and meters $10K in monthly model spend
- Built a pre-release test suite with **GitHub Actions**, expanding coverage from 5 smoke tests to 159 tasks so tool regressions were caught before release
- Established live monitoring in **Langfuse**, adopted by all 9 engineers, tracking per-session latency and failures and routing each failure to the engineer responsible
- Reduced wasted model spend by stopping runaway agent loops and using an **LLM judge** to flag unnecessary tool calls

**YOYO MARKETS** **New York, NY (Remote)**

***Software Engineer*** ***Aug. 2025 - Dec. 2025***

- Automated manual research and editorial work with a **Python** and LLM generation pipeline, producing **50+ markets per week at 5x the hand-curated rate**
- Developed an **asynchronous Python** scraper tracking 150+ high-signal Twitter accounts in real time, feeding that pipeline **3x the signal volume** of public trend aggregators
- Devised a ranking step that uses pairwise LLM comparisons and Elo ratings to screen 200+ candidate markets per cycle and send the strongest to founder review
- Shipped a **Next.js** review dashboard on **Vercel** that founders use daily, cutting **publishing turnaround from 30 minutes to under 5 minutes**

## RESEARCH AND PROJECTS

**EFFICIENT LLMS** | *PyTorch*, *CUDA*, *LoRA* **Jan. 2026 - May 2026 | USC**

- Converted **Llama 3.2-1B** into a **Mixture-of-Experts** model with **LoRA** adapters and kept held-out quality level with the original, perplexity **9.60 versus 9.66**
- Quantized weights from **FP16** to **INT4**, cutting model size **61%** while keeping generated answers coherent

**POWER GRID DISPATCH MODEL** | *Python* **Jan. 2026 - May 2026 | USC**

- Engineered a **learning-to-optimize** pipeline for New York's grid, where a model learns generation costs from public load, price, and outage data and an optimizer sets the hourly plan
- Cut the extra cost above the cheapest possible hourly plan **by 47%** versus the old schedule by adding transmission limits and a reserve margin

**MATH REASONING FINE-TUNE** | *Python*, *LoRA* **Aug. 2025 - Dec. 2025 | USC**

- Trained **Qwen2.5-1.5B** with **LoRA** on 1,000 grade-school math problems to write its reasoning before the answer
- Rebuilt the grader until a correct final answer **scored consistently**, then implemented a **GRPO** trainer that rewards a correct, clearly reasoned solution among 4 candidates

**WEENIX (A UNIX-LIKE OS KERNEL)** | *C*, *x86*, *QEMU* **Aug. 2025 - Dec. 2025 | USC**

- Constructed a Unix-like kernel in **C**, with process scheduling, a file system, and virtual memory
- Integrated the three parts in **QEMU** so processes could be scheduled, files created and removed, and a copied process shared memory until one side wrote to it

**POSTGRESQL BUFFER MANAGER** | *C*, *PostgreSQL* **Jan. 2025 - May 2025 | USC**

- Replaced **PostgreSQL**'s default memory rule with a first-in, first-out rule that discards the oldest unused data

- Benchmarked its performance against PostgreSQL's default implementation and found the simpler rule **matched its speed** at 1MB of memory

## EDUCATION

**UNIVERSITY OF SOUTHERN CALIFORNIA** **Los Angeles, CA**

*M.S. Computer Science | GPA: 3.6/4.0* *Sep. 2024 - May 2026*

Relevant Coursework: ML Accelerators, Operating Systems

**UNIVERSITY OF MICHIGAN, ANN ARBOR** **Ann Arbor, MI**

*B.S.E. Data Science | GPA: 3.7/4.0* *Sep. 2020 - May 2024*